



Your experiences are your expertise.

Carlos Cardenas-Iniguez, PhD

Assistant Professor of Clinical Population and
Public Health Sciences
Division of Environmental Health, Keck SOM, USC
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Deficit-based Models

Lack of Preparedness

Inadequate familial support

Cultural Mismatch

Inadequate social support

Not as engaged



Community Cultural Wealth Model (Tara Yosso, 2005)

Aspirational Capital

Is the ability to maintain hope and dreams for the future in the face of real and perceived barriers.

Linguistic Capital

Is the ability for students to develop communication skills through various experiences.

Familial Capital

Refers to the social and personal human resources in the environment, drawn from their extended familial networks (immediate, extended, and chosen).



Navigational Capital

Refers to skills and abilities to navigate “social institutions,” including educational spaces.

Resistance Capital

Developing knowledge and skills through challenging inequality

Social Capital

Networks within the community offering resources and support

The journey so far:

- **BS in Brain and Cognitive Science, MIT**
 - Technical Assistant, PI: John Gabrieli
- **MA and PhD In Psychology (Integrative Neuroscience), University of Chicago**
 - PI: Marc Berman
 - Certificate in Higher Education Administration
 - T32 in Health Services Research, AHRQ
- **Teaching Fellow in the Social Sciences, Psychology, University of Chicago**
- **Postdoctoral Fellow, Department of Population and Public Health, Keck School of Medicine, University of Southern California**
 - PI: Megan Herting
 - T32 in Environmental Genomics, NIEHS
- **Assistant Professor of Clinical Population and Public Health, Keck School of Medicine, University of Southern California**



My journey

Carlos Cardenas-Iniguez, PhD (he/him/his)



Birth!



San Diego

K-12



BS

Cambridge, MA



PhD

Chicago, IL



Postdoc

Los Angeles, CA

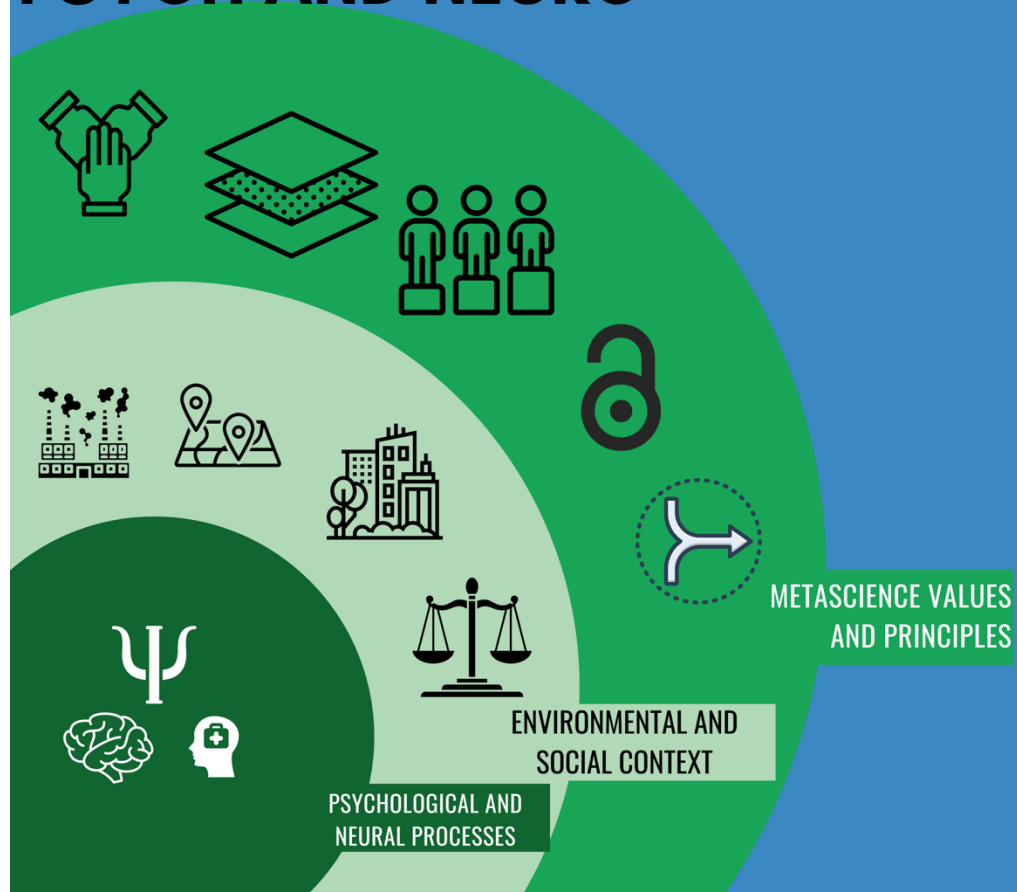


Keck Medicine of **USC**

Faculty



A MULTILEVEL FRAMEWORK FOR ENVIRONMENTAL FACTORS IN PSYCH AND NEURO

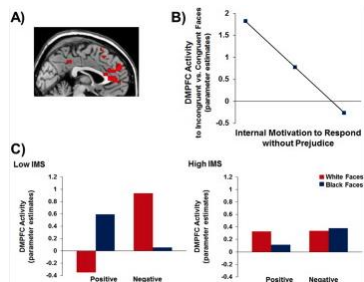


Investigation of basic processes in social and cognitive neuroscience related to language processing, person perception, and attention to social agents.

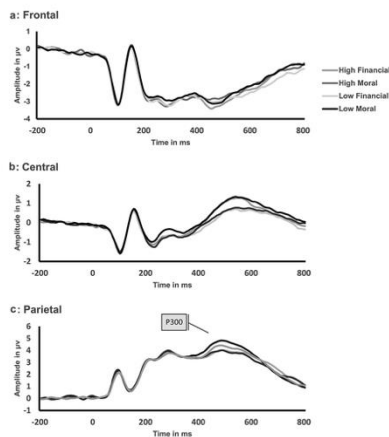
- Social Neuroscience**

Gyurovski et al., 2017, Social Neuroscience; Cacioppo et al., 2015, Cognitive Neuroscience; Li et al., 2015, NeuroImage.

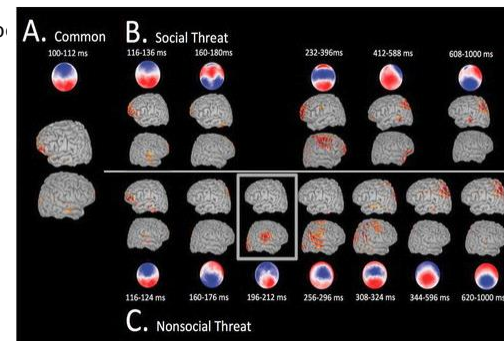
- Language**



Psychologia; Wu et al., 2010. A



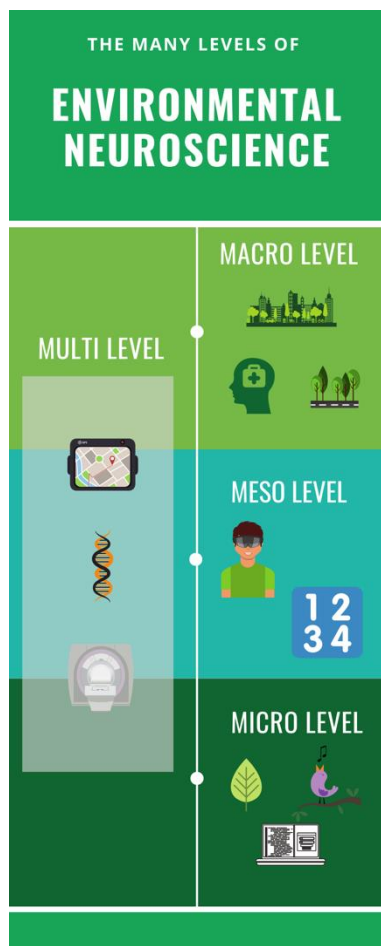
Li et al., 2009, Association



Li et al., 2015

Cacioppo et al., 2015

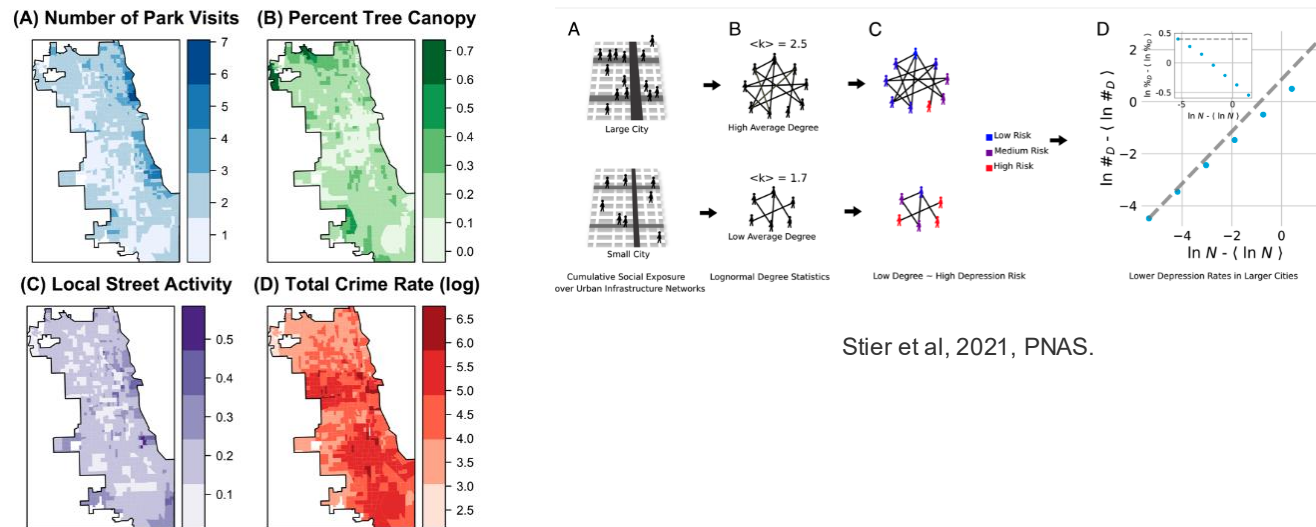
Gyurovski et al., 2017



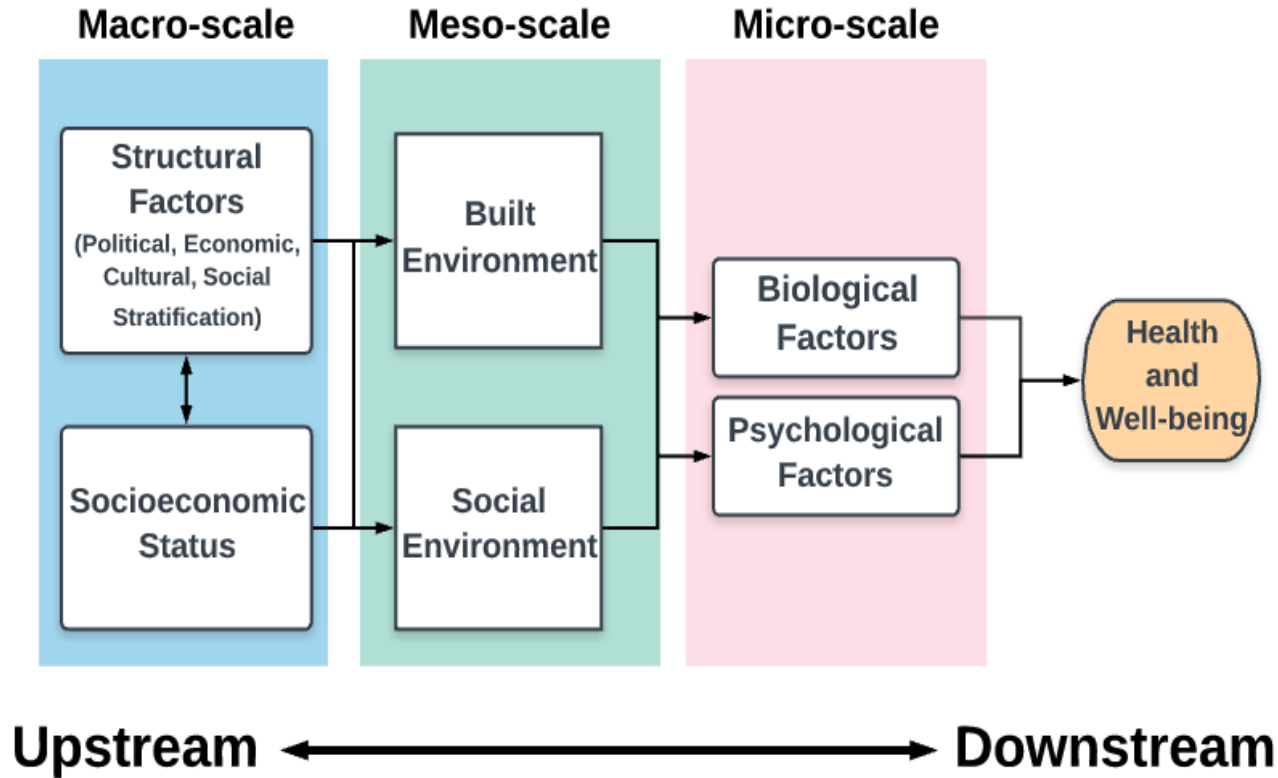
Environmental neuroscience approach to investigate role of urban context

A key feature of environmental neuroscience is the rigorous quantification of environmental features that affect the brain and subsequent behavior.

In addition, environmental neuroscience considers factors that vary across multiple temporal and spatial scales that interact to produce behavior. (Berman, Stier, & Akcelik, 2019; Berman et al., 2019)



Stier et al, 2021, PNAS.



Cardenas-Iniguez, C. (2019). *A Critical Examination of the Impact of Social Stratification and the Environment on Brain Development and Mental Health* [Ph.D., The University of Chicago].
<https://www.proquest.com/docview/2314066906/abstract/BEFD592F33ED47DBPQ/1>

Frameworks of Inspiration



Environmental Neuroscience



Social Epidemiology



Critical Neuroscience



**Intersectional and
Critical Race Theory**



The situation:

**Not many grad students of
color.**

**Not much support for grad
students.**

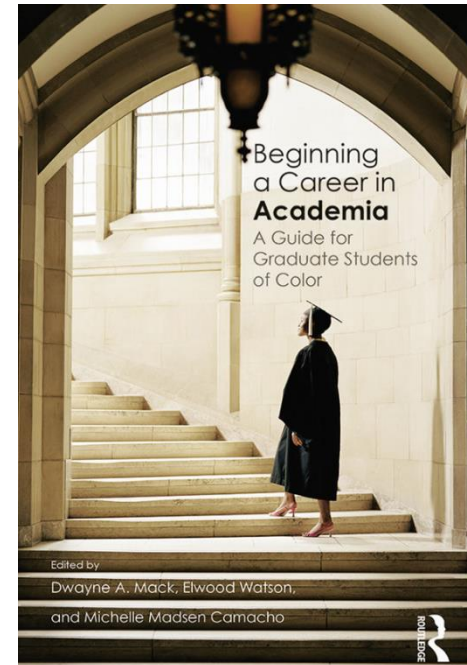
We secured an innovation grant!



UCHICAGO GRAD

**Social Sciences Minority Graduate Student
Academic Mentoring & Professionalization Program
Academic Year 2015-2016**

- Program for all 1st and 2nd years in the Social Sciences Division that were recognized as underrepresented minorities.
- Students met once a month for dinner workshops focusing on demystifying the hidden curriculum
- Were paired with a senior grad student in their department (money for lunches)
- Optional reading club





UChicagoGRAD



A network of people to support you.

At The University of Chicago, there are a number of offices committed to supporting students from underrepresented and marginalized backgrounds. Contact any of the offices listed here to find out about more about their resources and services.



GRAD POC PRE-ORIENTATION MASH-UP 2018

THURSDAY, SEPTEMBER 20

Transitioning to Graduate and Professional School

10:30am-Noon @ UChicagoGRAD

Welcome, plus overview of UChicagoGRAD
Current graduate student panel

LUNCH WILL BE PROVIDED AT NOON

Connecting for Success

1:30-3pm @ UChicagoGRAD

Creating and refining your elevator speech (with Michael O'Toole from GRAD Talk)
Speed networking event

FRIDAY, SEPTEMBER 21

Finding Your Folks and Setting Up Your Crew

10:30am-Noon @ UChicagoGRAD

Graduate diversity and inclusion resource fair

LUNCH WILL BE PROVIDED AT NOON

South Side and You

1:30-2pm @ UChicagoGRAD

Learn about "Grads on the Ground" program with Chris Huff from the Office of Civic Engagement

CI+I/OMSA Mixer for New and Current Graduate Students

6-8pm @ Center for Identity + Inclusion

5710 S Woodlawn Ave.
Food and Refreshments Provided



**RSVP via GRAD Gargoyle
Events > GRAD Events > Sign up**

Creation of Graduate Diversity Advisory Board

Diversity Advisory Board

PRIMARY PURPOSE

To provide input to [UChicagoGRAD](#) and other graduate-serving institutions to develop targeted programs and resources related to diversity in graduate education and experience.

THIS YEAR

Main focus on current underrepresented racial/ethnic minority graduate students

Focus on PhD Students

Meet three times a quarter

Stipend for Participation

Goals of DAB

Creating
Connections with
Academic Divisions
and Professional
Schools

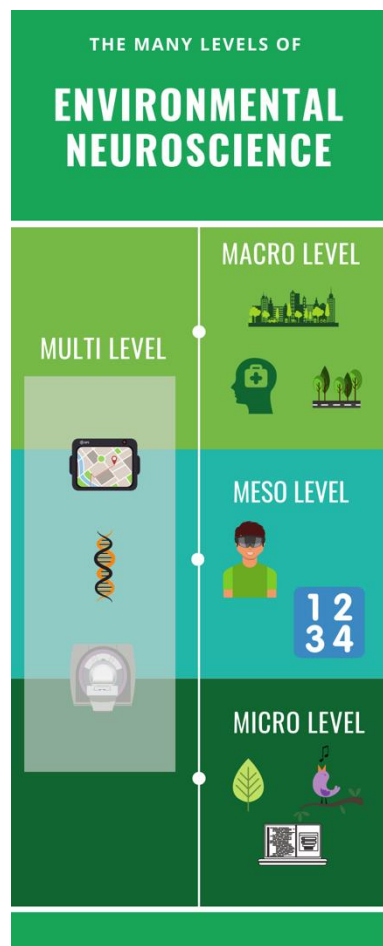
Centralization of
Graduate Programming
for Underrepresented
Minority Students
(URM)

Valuation of
Minority Research
and “Invisible
Diversity Labor”

Diversity Advisory Board

Current DAB Initiatives

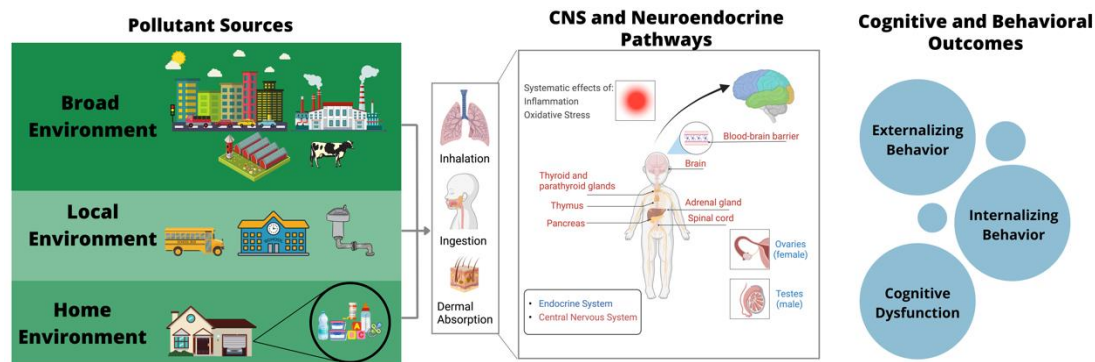
1. Transcending Boundaries Research Symposium (Jasmine)
2. Race and Pedagogy Working Group (Eilin)
3. DAB Townhall with Grad Students of Color (Camille)
4. Website and E-Newsletter (Carlos)



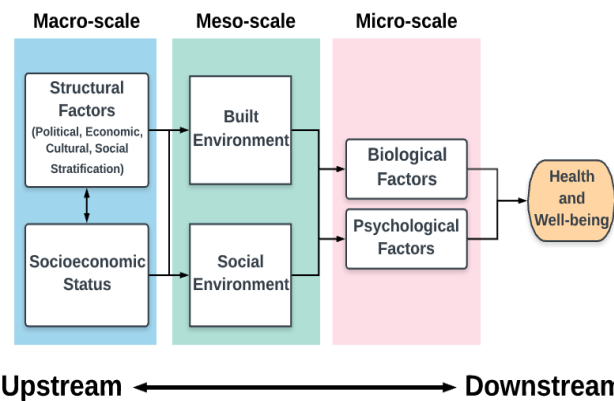
From Berman, Cardenas-Iniguez, & Meidenbaier, 2019.

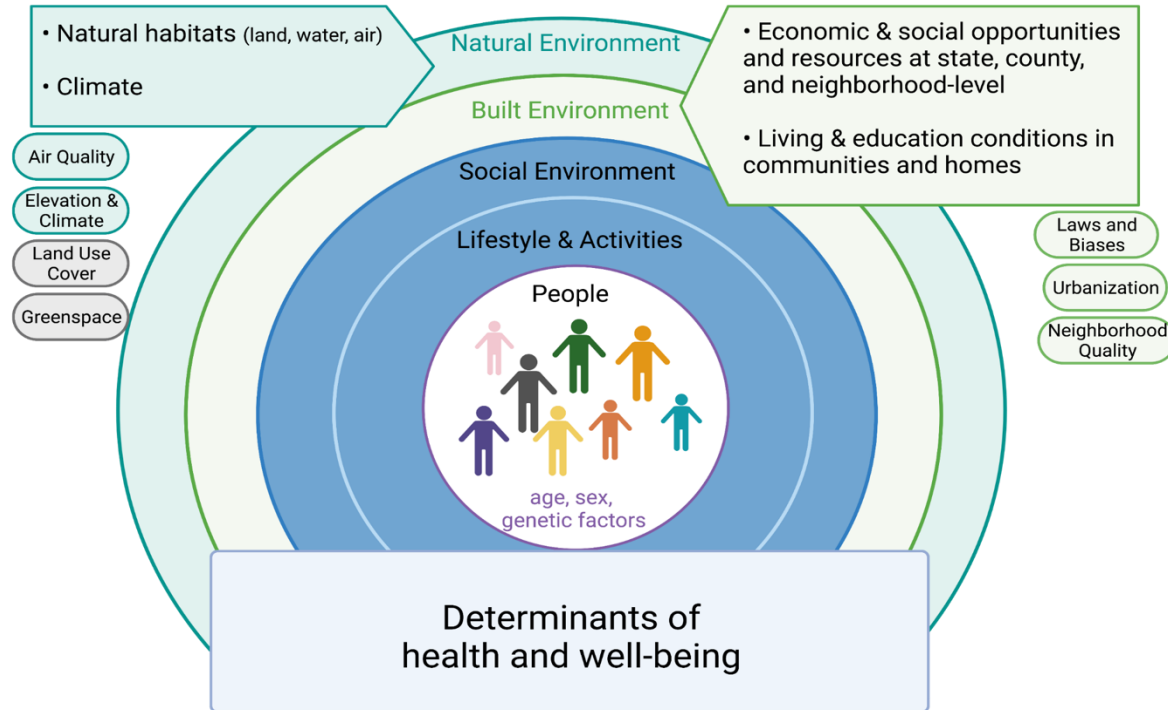


Environmental neuroscience approach to investigate The role of urban context



Cardenas-Iniguez, C., Burnor, E., Herting, M.M. Neurotoxicants, the developmental brain, and mental health. *Biological Psychiatry: Global Open Science*. (In Press)





Megan Herting, PhD
Associate Professor
Keck School of Medicine, USC



Adolescent Brain Cognitive Development[®]

Teen Brains. Today's Science. Brighter Future.

A longitudinal study of about 10,000 children from ages 9-10 through early adulthood to assess factors that influence individual brain development trajectories and functional outcomes

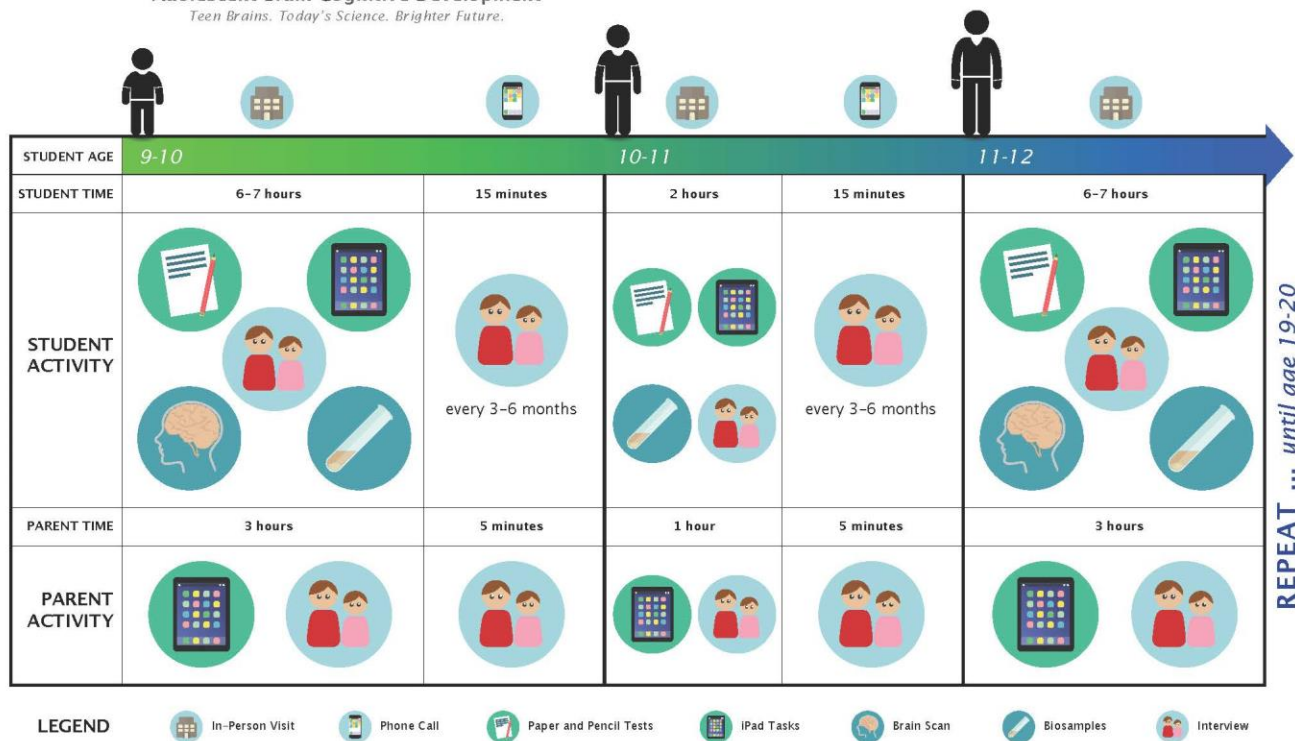


Adolescent Brain Cognitive Development®

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ABCD Study

TIMELINE OF EVENTS



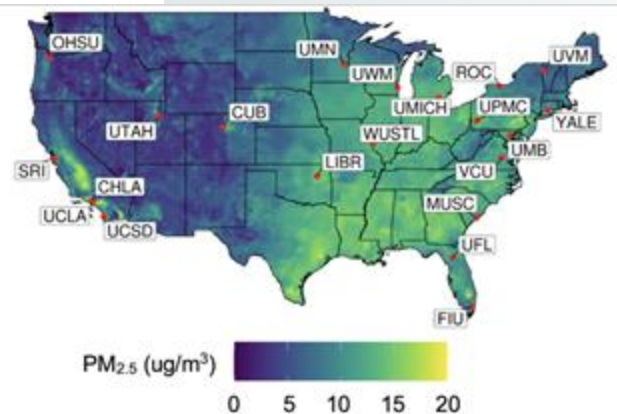


Adolescent Brain Cognitive Development®

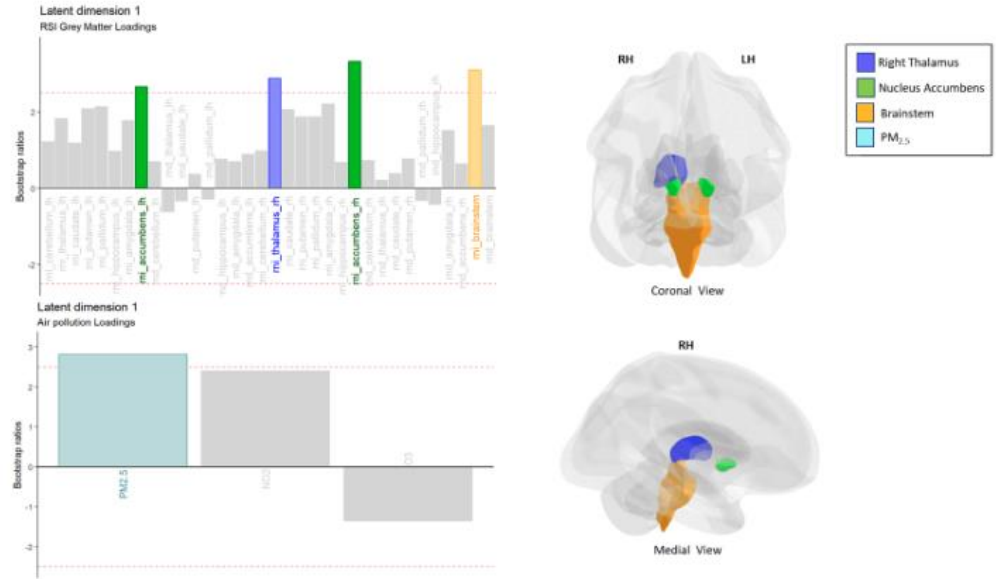
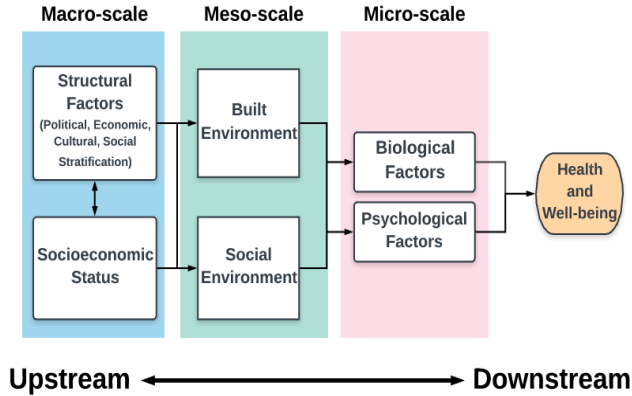
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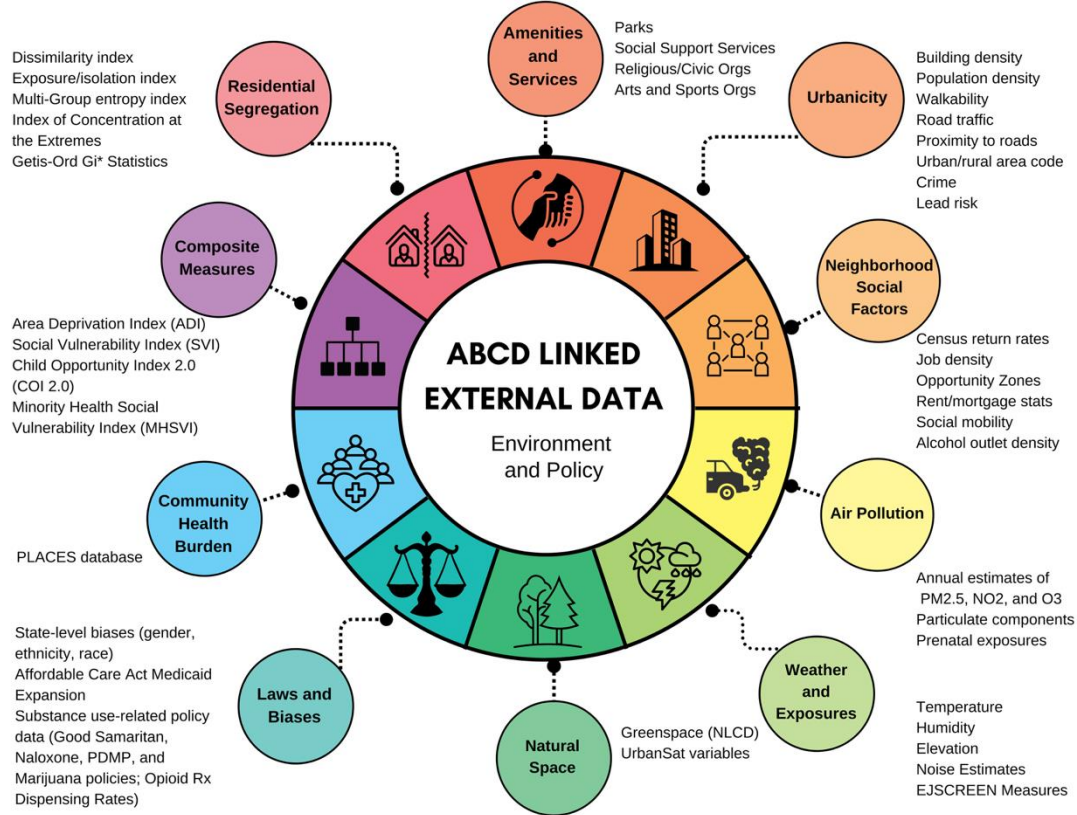
Adolescent Brain Cognitive Development (ABCD) Study



AIR POLLUTION EFFECTS ON SUBCORTICAL STRUCTURES



Sukumaran, K., Cardenas-Iniguez, C., Burnor, E., Bottenhorn, K. L., Hackman, D. A., McConnell, R., Berhane, K., Schwartz, J., Chen, J.-C., & Herting, M. M. (2023). Ambient fine particulate exposure and subcortical gray matter microarchitecture in 9- and 10-year-old children across the United States. *iScience*, 26(3), 106087. <https://doi.org/10.1016/j.isci.2023.106087>



Link to Paper

Cardenas-Iniguez, C., Schachner, J. N., Ip, K. I., Schertz, K. E., Gonzalez, M. R., Abad, S., & Herting, M. M. (2024). Building Towards an Adolescent Neural Urbanome: Expanding Environmental Measures using Linked External Data (LED) in the ABCD Study. *Developmental Cognitive Neuroscience*, 101338. <https://doi.org/10.1016/j.dcn.2023.101338>

A MULTILEVEL FRAMEWORK FOR ENVIRONMENTAL FACTORS IN PSYCH AND NEURO



Mentoring the next generation!



The Leadership Alliance



*Advancing Chicanos/Hispanics
& Native Americans in Science*



Científico Latino

Changing the next generation of STEM

This article is part of the Research Topic
Authentic Justice, Equity, Diversity and Inclusion in the
Neurosciences
[View all 8 Articles >](#)

A Developmental Approach to Diversifying Neuroscience Through Effective Mentorship Practices: Perspectives on Cross-Identity Mentorship and a Critical Call to Action

 Tanisha G. Hill-Jarrett^{1*},  Rowena Ng^{2,3*},  Carlos Cardenas-Iniguez⁴,
 Jemima Akinsanya⁵,  Ismary Blanco⁶,  Johnathan Borland⁷, James S. Brown⁸,
 Tameka Clemons⁹,  Adriana K. Cushnie⁷,  Jacqueline Garcia¹⁰, Brianna George¹¹,
 Cera W. Hassinan^{12,13},  Timothy J. Hines¹⁴, Dan Landayan¹⁵,  Taylor A. McCorkle¹⁶,
Katherine R. Meckel^{17,18}, Mariajose Metcalfe¹⁹, Samantha A. Montoya⁷, Deborah K. Rose²⁰ and
Desmond R. Warren²¹

C) Recommendations for Cross-Identity Mentorship Relationships

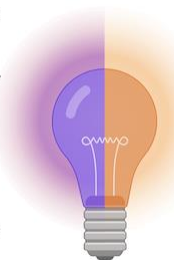
Mentors

Do not just be a manager, be an ally and sponsor. Acknowledge power imbalances in relationship.

Make expectations known, consider mentee's interests, listen to understand.

Connect on shared or similar lived experiences. Embrace differences, focus on strengths, and encourage authenticity.

Acknowledge and respect mentee's personhood outside of academia. Create and maintain a culture of communication and respect.



Mentees

Remember your identities and experiences are valid.

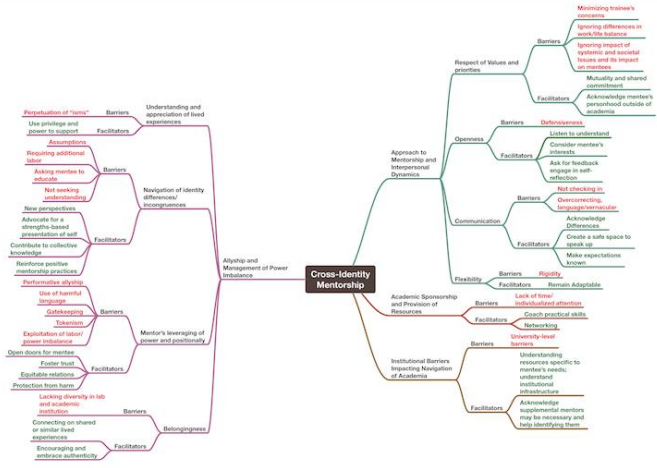
Communicating your needs is important.

Remember intra- and inter-personal factors when communicating with your mentor(s).

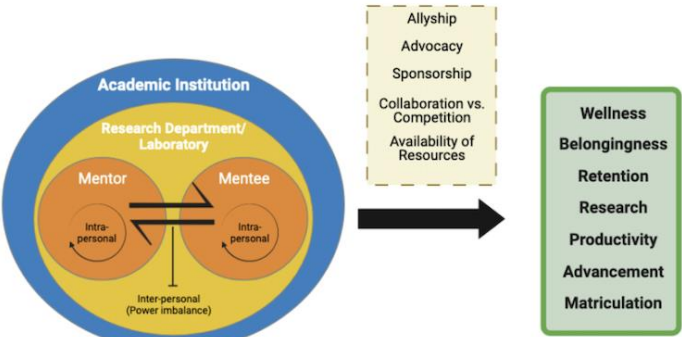
Ask for feedback and clarification of expectations.

Embrace a diverse multi-mentor network of support.

A)



B)





Radically reframing studies on neurobiology and socioeconomic circumstances: A call for social justice-oriented neuroscience



E. Kate Webb^{1,2,3†}



Carlos Cardenas-Iniguez^{4†} and



Robyn Douglas^{5†}



Kate Webb, PhD
McLean/Harvard Med
T/X: @_brainstorm_12



Robyn Douglas, MA
Texas A&M
T/X: @RobynDD_

RECOMMENDATIONS FOR THE RESPONSIBLE USE AND COMMUNICATION OF RACE AND ETHNICITY IN NEUROIMAGING RESEARCH

STUDY DESIGN AND CONCEPTUALIZATION

- 1 Race is socially constructed, with no biological basis.
- 2 Provide operationalization of race and ethnicity.
- 3 Avoid race and ethnicity as proxies, measure variables directly.

ANALYSIS AND IMPLEMENTATION

- 4 Report race and ethnicity of study population. Also ask: Why are you including race and ethnicity in models?
- 5 Race is not genetic ancestry.

RESEARCH CULTURE AND PRACTICES

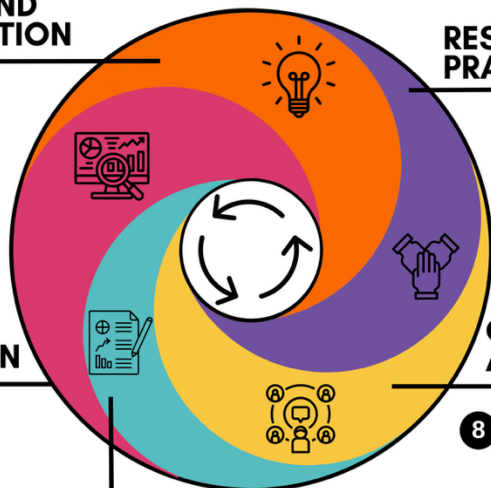
- 9 Develop a lab plan for responsible use of race and ethnicity.
- 10 Embrace that responsible use of race and ethnicity requires continuous intention and engagement.
- 11 Challenge standard conventions in the field related to minoritized groups.

COMMUNICATION AND DISSEMINATION

- 8 Champion strength-based models and prioritize the inclusion of minoritized voices.

APPLICATION AND INTERPRETATION

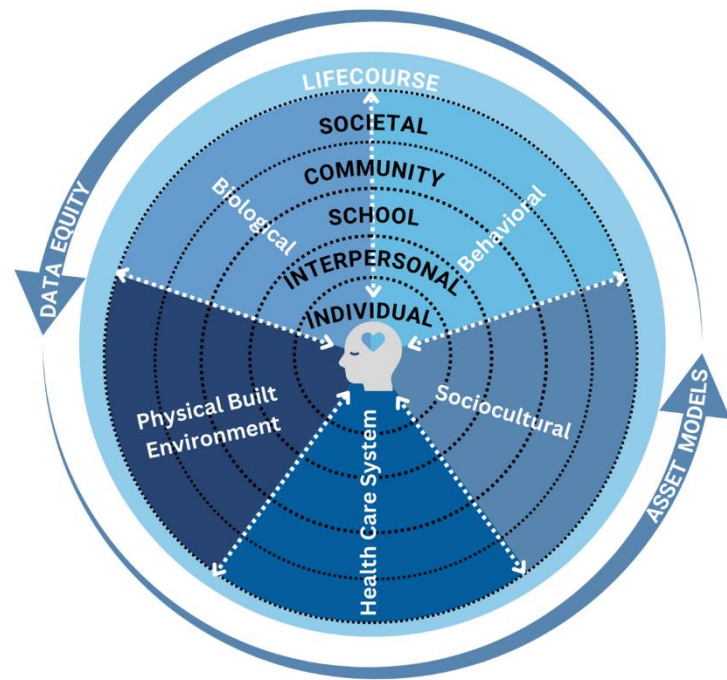
- 6 Evaluate biases in representation and participation, measurements, and generalizability.
- 7 Interrogate upstream causes instead of "race and ethnicity effects".



Marybel Robledo Gonzalez, PhD
Assistant Professor
The Ohio State University
College of Medicine

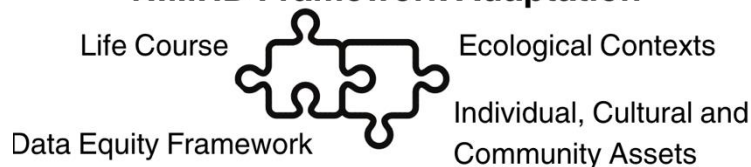


Cardenas-Iniguez, C., & Gonzalez, M. R. (2024). Recommendations for the responsible use and communication of race and ethnicity in neuroimaging research. *Nature Neuroscience*, 1–14. <https://doi.org/10.1038/s41593-024-01608-4>



People are embedded in multiple contexts and at various levels!

NIMHD Framework Adaptation



Marybel Robledo Gonzalez, PhD
Assistant Professor
The Ohio State University
College of Medicine





My journey

Carlos Cardenas-Iniguez, PhD (he/him/his)

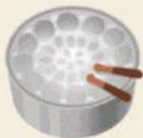


Birth!



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Postdoc

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Keck Medicine of **USC**

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